

Prices, Allocation, and Demand

Show your reasoning, not only the final answer. You may use without proof that the bipartite matching polytope is integral, so the LP relaxation has an optimal integral solution.

Problem 1: Can prices decentralize an efficient assignment?

Let $\mathcal{B} = \{1, \dots, n\}$ be a set of buyers and $\mathcal{I} = \{1, \dots, m\}$ a set of indivisible items. Buyer i has value $v_{ij} \geq 0$ for item j and wants at most one item; each item can be assigned to at most one buyer. Receiving no item gives utility zero. Our first goal is to find an allocation that maximizes total value, or *social welfare*.

- Introduce allocation variables x_{ij} and formulate the welfare-maximizing assignment problem as a linear program. Explain the buyer and item constraints. Why does integrality of the bipartite matching polytope allow us to solve an LP rather than an integer program?
- Derive the dual LP from scratch, including all sign constraints. Denote the dual variable for buyer i by u_i and the dual variable for item j by p_j . Rewrite the dual constraints in the form

$$u_i \geq v_{ij} - p_j,$$

and explain why u_i and p_j can be interpreted as buyer utilities and anonymous item prices.

- Write all complementary-slackness conditions for the primal and dual. Give an economic interpretation of each condition. Emphasize the edge condition

$$x_{ij} > 0 \implies u_i + p_j = v_{ij},$$

which will be the key to decentralization, as well as what $u_i > 0$ implies about buyer i and what $p_j > 0$ implies about item j .

- Let x be an integral primal optimum and (u, p) a dual optimum. Prove that every assigned buyer receives an item maximizing their nonnegative net surplus. In particular, if buyer i is assigned item j , show that

$$v_{ik} - p_k \leq u_i = v_{ij} - p_j \quad \text{for every item } k.$$

Explain why this proves that the efficient assignment x can be decentralized at prices p : every buyer is willing to accept the item assigned to them. If buyers have ties, must every arbitrary selection from their demand sets form a feasible assignment?

- Apply the general result to the following instance:

	A	B	C
1	10	8	3
2	9	5	7
3	4	9	8

Find a welfare-maximizing assignment and its total value. Then find one optimal dual solution (u, p) and verify directly that its item prices support your assignment. Ties are allowed: the assigned item need only be weakly utility-maximizing.

- (f) Verify that each of the price vectors

$$(p_A, p_B, p_C) = (5.8, 4, 3.7) \quad \text{and} \quad (p_A, p_B, p_C) = (5.5, 4, 3.4)$$

strictly supports the same efficient assignment. Compare seller revenue under the two price vectors. Do all supporting price vectors give the seller the same revenue? What does this show about the distinction between prices that decentralize an efficient allocation and the seller's revenue objective?

Problem 2: How does a distribution of values generate demand?

There are n potential buyers. Their values $V_1, \dots, V_n$ are independent and identically distributed according to a continuous CDF F with density f . Each buyer purchases one unit exactly when $V_i \geq p$.

- (a) Let $X_i(p) = \mathbf{1}\{V_i \geq p\}$ and let realized demand be $Q_n(p) = \sum_{i=1}^n X_i(p)$. Find the distribution, mean, and variance of $Q_n(p)$. Denote its expected demand by $d(p) = \mathbb{E}[Q_n(p)]$. Also compute its standard deviation and relative standard deviation (coefficient of variation),

$$\frac{\sqrt{\text{Var}(Q_n(p))}}{\mathbb{E}[Q_n(p)]}.$$

For a fixed purchase probability in $(0, 1)$, identify the orders of $d(p)$ and $\text{sd}(Q_n(p))$ in n . How do relative fluctuations scale with n ?

- (b) Let random total revenue be $R_n(p) = pQ_n(p)$. Show that expected total revenue is

$$r_n(p) := \mathbb{E}[R_n(p)] = np(1 - F(p)).$$

Define the per-buyer expected revenue curve

$$r(p) := \frac{r_n(p)}{n} = p(1 - F(p)).$$

Find an optimal price when values are (i) uniform on $[0, m]$, restricting attention to $p \in [0, m]$, and (ii) exponential with rate λ , with $p \geq 0$. For each case, determine whether revenue is concave on the stated price domain.

- (c) Reparameterize price by the purchase quantile

$$q = 1 - F(p), \quad p(q) = F^{-1}(1 - q),$$

and define the *per-buyer revenue curve* $r(q) = qp(q)$. Derive $r(q)$ for the uniform and exponential distributions and show that both curves are concave for $q \in (0, 1)$. Explain why $r(p)$ and $r(q)$ are the same expected-revenue function written in two different coordinates.

- (d) Define the virtual value

$$\phi(v) = v - \frac{1 - F(v)}{f(v)}.$$

Show that $r'(q) = \phi(p(q))$. Explain why concavity of r is equivalent to virtual value being nondecreasing (under the regularity conditions used here).

- (e) The hazard rate is $h(v) = f(v)/(1 - F(v))$. Rewrite virtual value using the hazard rate and show that a nondecreasing hazard rate implies a nondecreasing virtual value. This is the implication “MHR $\Rightarrow$ regular.”

Problem 3: Pricing a congested service

This problem revisits the Calibration Day setting from the Queue Lab in Lecture 1. One robot arrives every two cycles at Robot City's only repair bay. Each repair requires three cycles, every robot has mission value 5, and delay costs 1 per cycle. When a robot arrives, let w be the current repair workload: the number of repair cycles already scheduled. A robot offered access price τ docks exactly when

$$5 - \tau - w \geq 0.$$

If the robot docks, the workload at the next arrival is $w + 1$; if it reroutes, it is $\max\{w - 2, 0\}$. Initially $w = 0$.

- (a) If the repair bay is idle, what is the largest access price the arriving robot will pay? More generally, state the largest nonnegative price that induces the robot to dock at workload w . What happens when $w > 5$?
- (b) Consider the policy that extracts each arriving robot's full current willingness to pay whenever a nonnegative price can induce it to dock. Write the workload sequence until it becomes periodic. Compute the operator's long-run average credits per cycle.
- (c) Now charge the fixed access price $\tau = 5$. Find the periodic workload sequence and the operator's long-run average credits per cycle.
- (d) Finally, charge 5 when $w = 0$ and 4 whenever $w > 0$. Find the periodic workload sequence and the operator's long-run average credits per cycle.
- (e) Compare the three policies. Explain why maximizing revenue from the current robot need not maximize long-run revenue. What scarce resource is the access price rationing? For each robot that docks, what is its contribution to city surplus if payments are treated as transfers?